

Data Science Project Proposal

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1. Summary

- The purpose of this project (actually a sub-project) is to satisfy the requirements for a College-level Data Science project, as well as to further my study of Machine Learning and Computer Science.

2. Introduction and Background

- The issue prompting this project is that while encoding and decoding CW/Morse is a relatively trivial task for *humans*, it is not so for a computer.
- Morse is an intricate sequence of a short and long tones, differentiated by and grouped by very precisely-timed intervals of relative silence. As such, each transmission shares common characteristics:
 1. Every observation has a tone of a specific frequency (Hz), accompanied by some amount of parasitic noise (whether the noise floor/"background noise", or man-made interference.
 2. A static pair of tonal emissions of a known length:
 - A dit is one measure of time $\leftarrow$ The standard measure μ
 - A da is 3μ measures of time.
 3. Strict voids between morse "characters", morse groups "letters", and groups of groups "words"
 - intra-character spacing of μ
 - Inter-character spacing of 3μ
 - Word/Character group spacing of 7μ

3. Methodology and Approach

- My planned approach is to use [a set of training data](#) from kaggle, which will be analyzed in R using various candidate libraries to conduct **Exploratory Data Analysis**, before proceeding to extracting the codes and processing the messages with libraries including but not limited to:
 - [Sonic Screwdriver Lib - CRAN](#)
 - [Seewave Lib](#)
- Alternatively, if the first proposed dataset proves insufficient to study, Creating some artificial data with the [already-existing implementation](#) that this app will support is a trivial task.
- Conceptually, I feel like there are several paths towards a solution:
 - Geometric Mean: By far the simplest and most straightforward method is to measure dits and dashes through establishing a cutoff threshold for "tone on/off" and using the [geometric/arithmetic mean](#) to continually evaluate what must be a dit/dash based on how the small bit of sound compares to this running calculation. This has been implemented on very minimal hardware such as an arduino, and it's admirable for its simplicity. I think this would be a good point to start the project, because it directly ties in with **evaluating and telling the story of the data**, which is a key deliverable for an Intro to Data Science course. From there, time permitting, perhaps the more advanced methods can be added to improve accuracy as needed.
 - Bayesian Probability: [Some implementations](#) have attempted to use Bayesian probability to estimate the *probability* that a given tone is a dit/da or even an entire word fits inside a given *dictionary* of words. This could potentially be used by itself, or as a type of post-processing after a simpler and lighter algorithm does the bulk of the heavy lifting. Many practical conversations are very formulaic and predictable, due to the nature of HAM radio; The typical conversation usually consists of a callsign, a brief exchange of location, reception clarity, contest details, type of equipment being used to transmit and receive with, and an end of transmission "goodbye." The small sample space makes this a good option - even more so in the case of a robot that's exchanging a highly formulaic message sequence such as:

I.e. even if a rather error-prone decoding algorithm is only right at determining, for example, 3/5 of the dits or da's with dependable accuracy, a Bayesian algorithm should be able to determine the intended word.
 - Clustering: There's promising literature that indicates clustering is very good for identifying [bird sounds](#) in audio recordings. Just as birds have a distinct sound, so does each letter in CW. Per the guidance of the largest CW enthusiasts club in the country, humans learn best by recognizing entire sounds, as opposed to decoding the sequence of dashes and dits on the fly.
- Machine Learning could be applicable in certain cases, particularly when normal analysis cannot predictably decipher a signal; given that morse can be transmitted across a range of tones (Hz) and speeds (WPM), there are more scenarios than would be efficient to account for by "hard coding" every possible variation. Additionally, since this data is gathered from a microphone, there's strong potential for interference (static and environmental noise), that may be impossible to remove with filtering techniques.

4. Timeline

Per the course outline, this project is to be managed in the **SCRUM agile project management** style, where we are encouraged as aspiring Data Scientists to continually *evaluate* and *learn through our experiences* with attention to our personal **skill stack and problem-solving abilities** as they relate to the task at hand.

Particularly, the project will be divided into four checkpoints:

1. **This proposal stage:** where the idea is pitched to a client or end-user.
2. **Literature Review:** wherein we evaluate other scholarly material to gain helpful insights. A core skill in good Science is having an adequate grasp of what others have tried before.
3. **Abstract:** Likewise, it is important that one's research is clearly articulated. An abstract helps other readers to efficiently determine whether our work is relevant to their studies.
4. **Presentation:** Finally, the conclusion of our study is presented to other scientists, students, colleagues, or more universally an end-user. Almost always done in a live "show-and-tell" style, this an opportunity to showcase a project and generate further interest to continue development or gain patronage for a future project (or to pass a class, in this case).

5. Evaluation

Evaluation will be conducted formally by the course instructor per the University's grading policy, the course syllabus, and the grading rubric as provided for each stage. In addition, less informal but no less important evaluation is conducted by the researcher, per the aforementioned agile framework.

Strictly in terms of measuring the success of any meaningful data extracted, the candidate Kaggle dataset mentions that the *Levenshtein distance* is the metric by which any candidate solution to their challenge is objectively evaluated. They provide this as a part of the dataset, so the outcome is measurable beyond grading.

6. Conclusion

Foremost, it is this project's intention to examine CW/Morse Code audio emissions gathered by microphone via digital signal processing from a Data Science perspective. Audio files themselves are data structures which contain meaningful data, as well as outlier data: static, or literal noise, which can be **cleaned, manipulated, analyzed, and visualized** in order to **extract meaningful insights and information:** an encoded message.

Ultimately, Data Science in this context furthers the parent program's ability to decode raw data from a microphone into actionable information that can either be displayed to an end-user and/or executed directly by another program in an application stack as part of a larger project. As such, if this project can meet its goals, it not only adequately demonstrates Data Science's intrinsic value in analyzing and understanding complicated information such as audio waveforms, but additionally showcases its value as an applied science to further other other fields, such as robotics, telecommunications, and machine learning, which are likewise key areas relating to the larger project.